Exercise – *A Mad Tea-Party*

"Take some more tea," the March Hare said to Alice, very earnestly. "I've had nothing yet," Alice replied in an offended tone, "so I can't take more."

Alice's Adventures in Wonderland *by Lewis Carroll (1832–1898)*

As Alice wanders through the whimsical world of Wonderland she comes across the March Hare and Mad Hatter engaging in their eternal tea party. She notices plenty of empty chairs and decides to join them. Her arrival is met with protest. "It is not very civil of you to sit down without being invited" proclaims the Mad Hatter dramatically, while the March Hare nods in agreement.

Yet Alice remains seated, fixing them both with a calm stare. Sensing her unyielding spirit, the Mad Hatter concedes: "Very well, your audacity shall earn you an opportunity to tilt the scales in your favor. The Queen of Hearts herself plagued us with a riddle unlike any we have faced before. You see, Her Majesty is orchestrating an exquisite Algolab exam for our fellow denizens to determine who will have the honor of participating in her croquet game. The exam will take place in a mystical mushroom field, organized in a grid, where each coordinate is either empty or contains a magical mushroom with a special ability, allowing denizens to write, debug, and submit their code. Each mushroom can be occupied by at most one denizen." The March Hare interrupted: "Denizens could cheat by looking at the answers of those sitting to their right, left, front right, or front left. Denizens cannot see the answers of those located directly in front of them or behind them. Thus, if a denizen occupies a mushroom on row i and column j, that is, with coordinates (i, j), they could cheat by looking at the answers of denizens at mushrooms $(i, j-1)$, $(i, j+1)$, $(i-1, j-1)$, and $(i-1, j+1)$." The Mad Hatter continues: "Cheating at an Algolab exam is despicable. Thus, the Queen demands that we find the maximum number of denizens that can be placed without any cheating being possible. Help us solve this riddle, and we should bother you no more. If you manage this feat, you will always have a place in our tea party."

Alice never backs down from a riddle and decides to help the Mad Hatter and March Hare determine the maximum number of denizens that can sit for the exam.

Input The first line of the input contains the number $t \leqslant 30$ of test cases. Each of the t test cases is described as follows.

- It starts with a line containing two integers n m, separated by a space, where $1 \leqslant n \leqslant 50$ and $1 \leqslant m \leqslant 50$ denote the number of rows and columns, respectively, of the grid that represents the mushroom field.
- The following n lines contain m characters each, where the j-th character of the i-th line denotes whether there exists a magical mushroom (denoted by 1) or not (denoted by 0) at position (i, j).

Output For each test case you should output the maximum number of denizens that can sit for the Algolab exam without any cheating being possible.

Points There are three groups of test sets. For each group there is also a corresponding hidden test set. Overall, you can achieve 100 points.

1. For the first group of test sets, worth 25 points, and for the corresponding hidden test set, worth 5 points, you may assume that $m = 2$.
2. For the second group of test sets, worth 25 points, and for the corresponding hidden test set, worth 5 points, you may assume that every mushroom has at most two other mushrooms in its eight neighboring grid cells.
3. For the third group of test sets, worth 30 points, and for the corresponding hidden test set, worth 10 points, there are no additional assumptions.

Sample Input

```
2
15 2
0 0
1 1
1 1
0 1
1 1
1 0
1 1
0 0
0 1
1 1
0 0
0 0
0 0
1 1
0 0
2 5
1 0 1 0 0
0 1 1 0 1
```

Sample Output

```
8
4
```